VetLink PWA - Technical Progress & Data Integrity Report

College of Computing Studies, Eulogio 'Amang' Rodriguez Institute of Science and Technology (EARIST)
Capstone Project: AI-Assisted Veterinary Skin Disease Decision Support System | Updated: November 6, 2026

1. Executive Summary & Current Completion Status

The VetLink clinical progressive web application has achieved an overall completion rate of **98%**. All clinical workflows, biometric quality evaluations, on-device and cloud neural inferences, database normalization, and audit trails are fully implemented, verified, and operational.

2. Module Implementation & Verification Matrix

Component / Module	Status	Key Technical Accomplishments
Database Normalization & Owners Table (3NF)	**100% Complete**	Activated owners table in MySQL and IndexedDB. Implemented Smart Auto-Linking: intake checks existing owners, links patient.owner_id, and auto-populates owner records without dual forms.
Data Privacy Compliance (RA 10173)	**100% Complete**	Enforced Data Minimization (Principle of Proportionality). Replaced intrusive street addresses with optional City/Barangay locality tracking for veterinary epidemiological disease surveillance (Scabies/Ringworm).
Demographics & Persistence Fix	**100% Complete**	Resolved screenshot bug: Weight in kg, Owner Name, Contact Phone, Locality, and Email are fully saved across IndexedDB, Edit Modals, and MySQL cloud sync schema.
Offline AI Hardening & Guard	**100% Complete**	Eliminated false 20.9% Pyoderma flat fallback. Configured Workbox precaching for 3.5 MB neural weights and WASM binary. Verified offline execution next day without internet.
Soft-Delete & Recycle Bin	**100% Complete**	Added is_archived schema flag. Implemented Archive and Restore actions in Patient Records, preventing accidental clinical record loss with full audit logging.
Audit Trail & Access Control	**100% Complete**	Created audit_logs table. Logs PATIENT_ADMITTED, PATIENT_EDITED, ARCHIVED, RESTORED, and RX_FINALIZED. Added real-time token/session rechecks preventing stale role bypasses.
Clinical Flow: Intake & Rx Modal	**100% Complete**	NewPatient form auto-resets on save/transition. Treatment page features formal confirmation modal redirecting clinicians back to /patients upon prescription finalization.
Cloud Sync & Conflict Resolution	**100% Complete**	Sync pipeline handles users, owners, patients, consultations, screeningResults, and auditLogs with version-based conflict resolution.

3. Thesis Defense Alignment (Relational Design & Data Privacy)

Third Normal Form (3NF) Relational Separation: VetLink implements a 1:M relationship between owners and patients. A single pet owner can register multiple animal patients over longitudinal visits, eliminating update anomalies and redundant client profiles.

Data Privacy Act of 2012 (RA 10173) Defense Script: *'In strict compliance with RA 10173 and the Principle of Proportionality, VetLink avoids collecting sensitive residential home addresses. Address data is collected on an optional City/Barangay level solely for veterinary epidemiological disease tracking to detect contagious dermatological clusters such as Sarcoptic Mange and Dermatophytosis.'*

4. Build Verification & Git Branch Deliverables

- **Production Build:** npm run build succeeded with 0 errors (31 precache entries, 14.15 MB total package).
- **Python Backend:** backend/main.py compiled and verified with py_compile.
- **Feature Branch:** feature/clinical-workflow-and-data-integrity containing all unified frontend, backend, schema, and documentation updates.